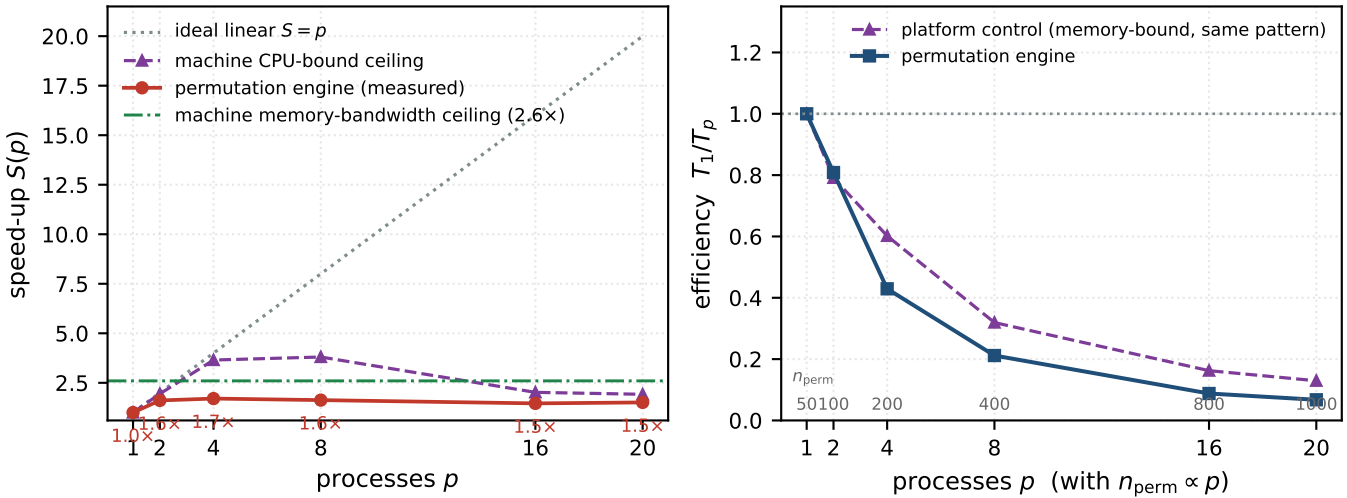
(a) Strong scaling vs. platform ceilings
ideal linear $S = p$
machine CPU-bound ceiling
permutation engine (measured)
machine memory-bandwidth ceiling (2.6×)
speed-up $S(p)$
processes p
1.0×
1.6×
1.7×
1.6×
1.5×
1.5×
(b) Weak scaling
platform control (memory-bound, same pattern)
permutation engine
efficiency T_1/T_p
n_{perm}
50
100
200
400
800
1000
processes p (with $n_{\text{perm}} \propto p$)